[2024-2025 春夏] 离散数学理论基础 quiz2

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1. (5% each)

   (1) Find the next larger permutation in lexicographic order after 2716543. _____

   (2) Find the next larger 4-combinations of the set $\{1,2,3,4,5,6,7,8\}$ after $\{1,3,5,8\}$. _____

   (3) Suppose $|A| = |B| = |C| = 100$, $|A \cap B| = |A \cap C| = |B \cap C| = 55$, and $|A \cup B \cup C| = 180$. There are _____ elements in $A \cap B \cap C$.

   (4) Find the value of $\binom{-0.5}{4}$. _____

   (5) How many relations are there on a set with n elements that are both reflexive and symmetric? _____

   (6) How many relations are there on a set with n elements that are neither symmetric nor anti-symmetric? _____

   (7) How many transitive relations are there on a set with 2 elements? _____

   (8) How many ways are there to distribute 8 balls into 5 boxes, if no box can be left empty, both the balls and boxes are labeled? _____

   (9) How many different ways can you put 9 balls in 9 boxes, if both the balls and boxes are labeled, and each box is either empty or contains exactly 3 balls? _____

   (10) How many different ways can you put 9 balls in 9 boxes, if the balls are identical and the boxes are labeled, and exactly 6 boxes are empty? _____

   (11) Find the coefficient of $x^5y^3z^2$ in the expansion of $(x+2y+z)^{10}$. _____

   (12) If $G(x)$ is the generating function for the sequence $\{a_k\}$, what is the generating function for the sequence

   $$a_0,\ 0,\ 2a_1,\ 0,\ 4a_2,\ 0,\ 8a_3,\ 0,\ 16a_4,\ 0,\ldots$$

   _____

2. (10%) A decimal number is called a "lucky number" if the sum of its digits equals 9. For example, 2025 is a lucky number (2+0+2+5=9).

   (1) How many lucky numbers are there between 1 and 100,000? _____

   (2) If the lucky numbers between 1 and 100,000 are sorted in increasing order, what is the position of 2025? _____

3. (10%) In the ordered pair (a,b), where a and b are both positive integers and the least common multiple of a and b is 2025, how many different ordered pairs satisfy these conditions?

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4. (10%) Solve the recurrence relation

$$a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3} + 1, \quad a_0 = 2,\ a_1 = 4,\ a_2 = 8.$$

5. (10%) Let $A = \{a_1, a_2, \ldots, a_8\}$ be a set consisting of 8 distinct positive integers, each satisfying $a_i \leq 35$. Prove that A must have two different disjoint subsets whose elements have the same sum.